

A Short Practical Introduction to Bayesian Analysis

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24–25 September 2026

A short worked example

The Bayesian toolkit

The Bayesian workflow

Working through the workflow: golf putting

Preparing for active learning: sleep deprivation

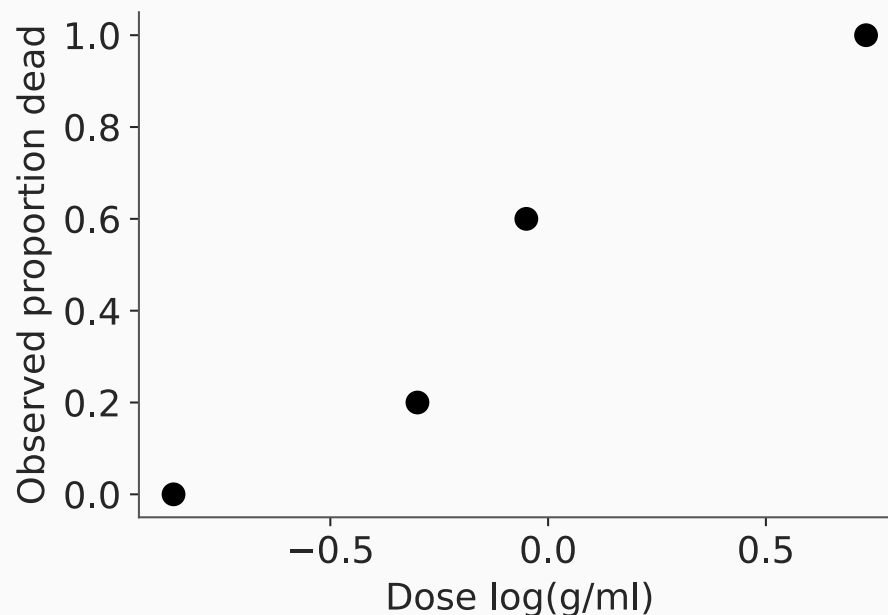
A short worked example

Dose response

Four doses, twenty animals

A toxin is given to rats at four doses, five rats per dose.

log dose	rats	deaths
-0.86	5	0
-0.30	5	1
-0.05	5	3
0.73	5	5

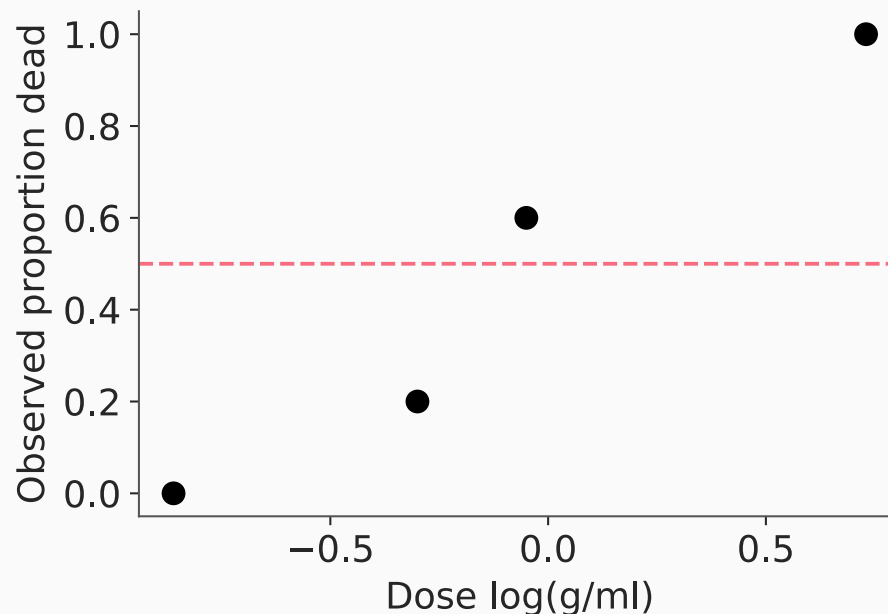


Data: Racine-Poon et al. (1986); Gelman & Vehtari, *Bayesian Workflow*, §3.5

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A generative model for mortality

$$y_j \sim \text{Binomial}(n_j, p_j)$$

$$\text{logit}(p_j) = \alpha + \beta x_j$$

$$\alpha \sim \text{Normal}(0, 5)$$

$$\beta \sim \text{HalfNormal}(5)$$

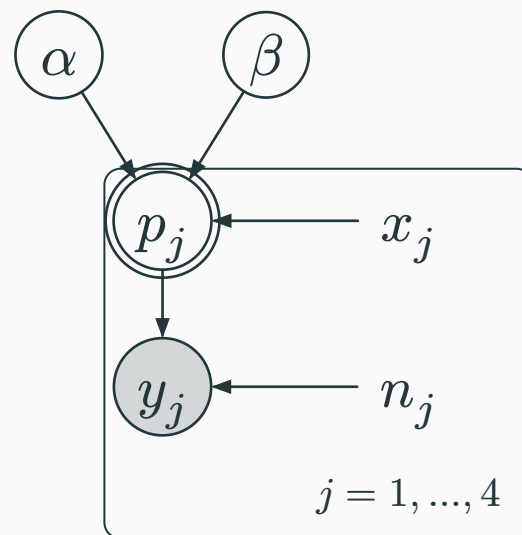
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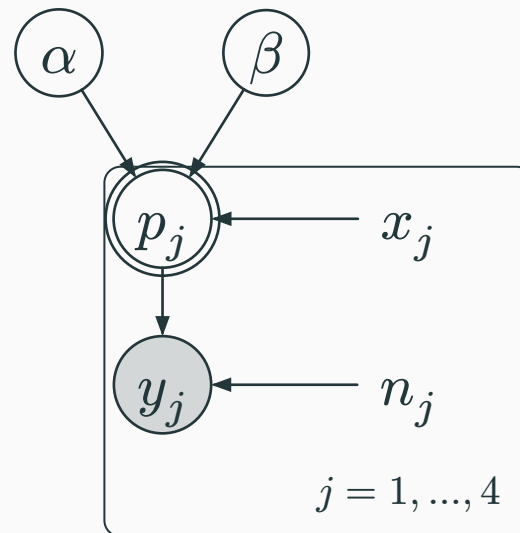
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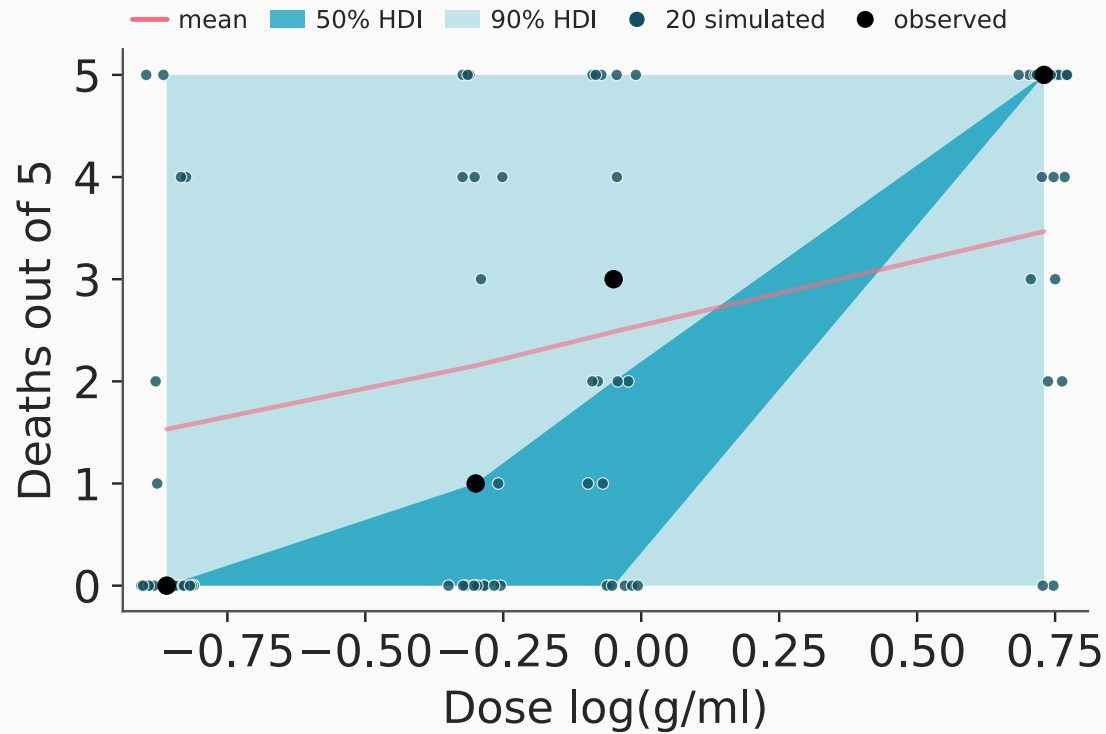
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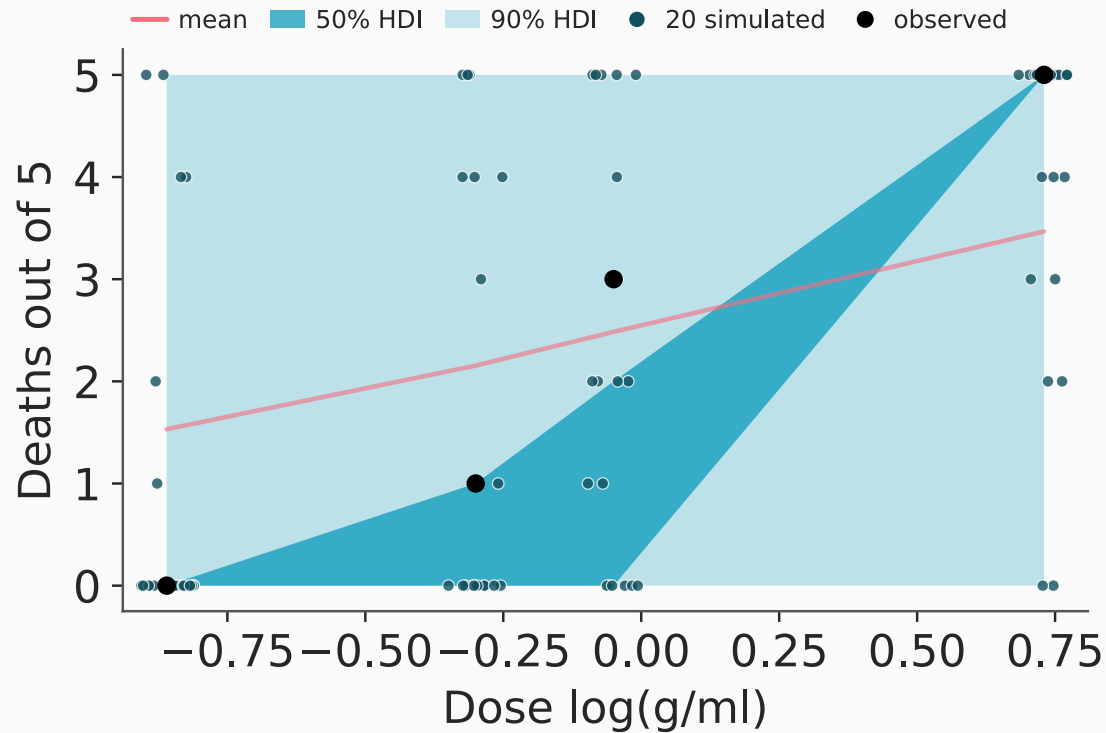
```
with pm.Model() as model:  
    alpha = pm.Normal("alpha", mu=0, sigma=5)  
    beta = pm.HalfNormal("beta", sigma=5)  
    logit_p = alpha + beta * dose  
    pm.Binomial("deaths", n=n, logit_p=logit_p, observed=deaths)
```

Priors: what could we have seen?



Prior predictive data should not

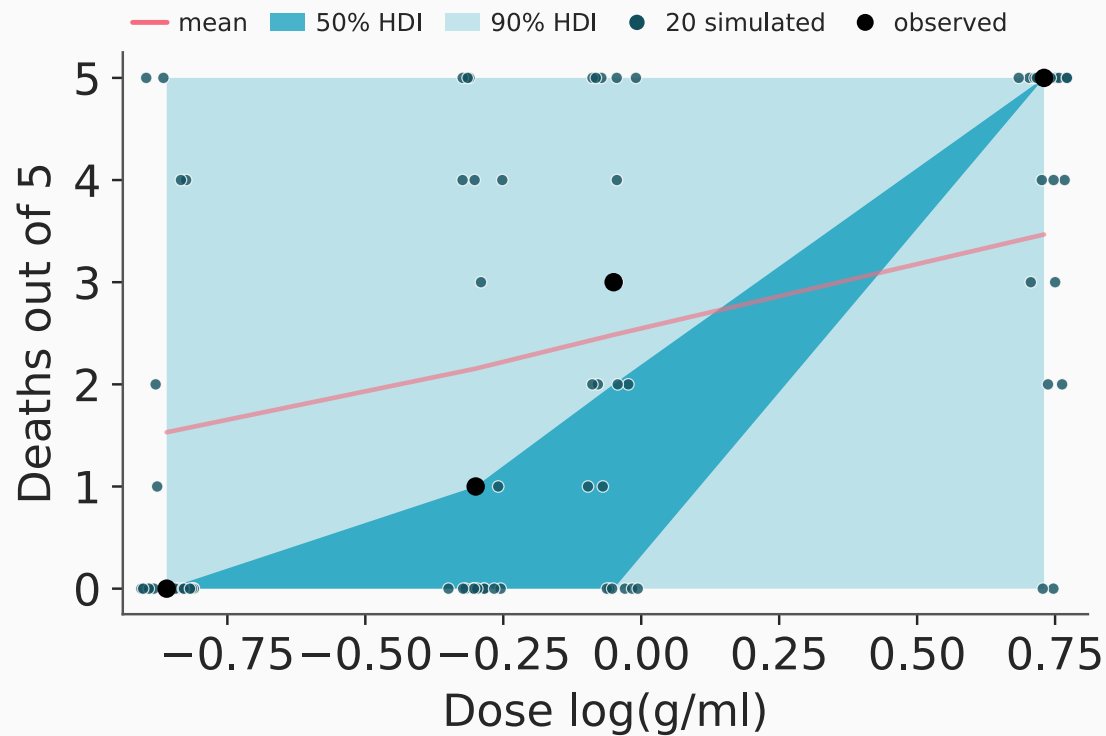
Priors: what could we have seen?



Prior predictive data should not

- violate common sense

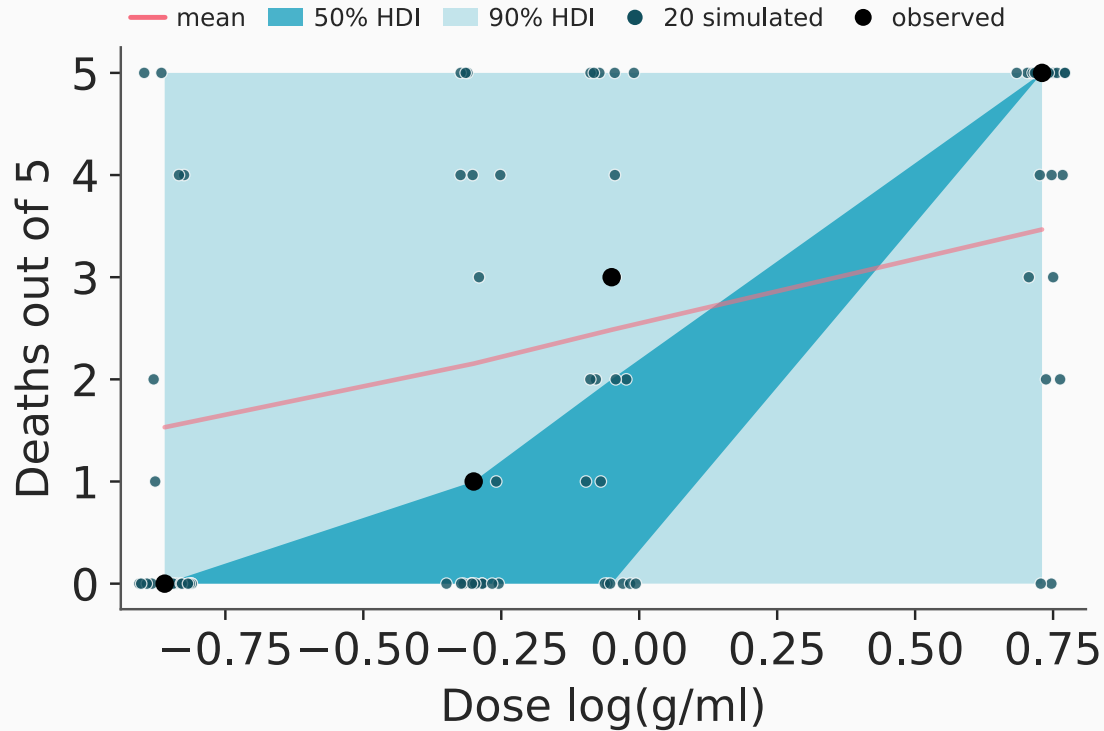
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- make our data highly unlikely

Priors: what could we have seen?



Prior predictive data should not

- violate common sense
- make our data highly unlikely
- confine the model to our data

Sampling and diagnostics

Fitting the model means drawing samples from the posterior distribution.

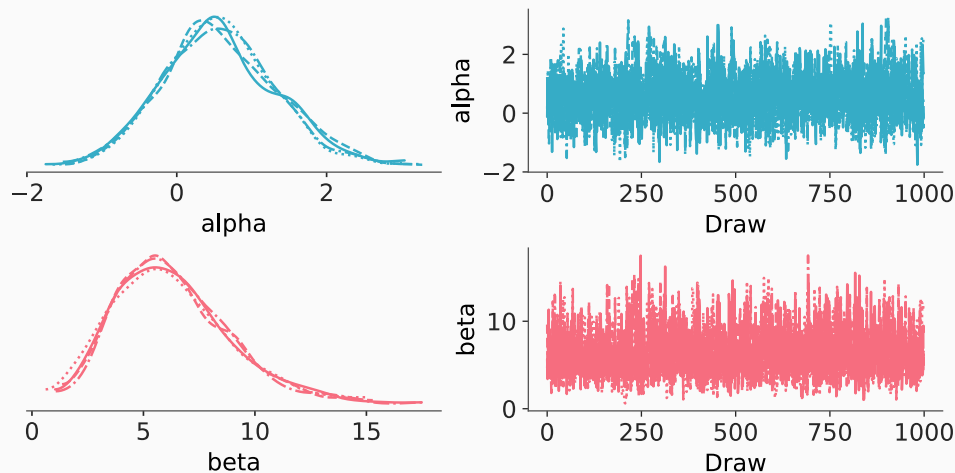
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with model:  
    idata = pm.sample(draws=1000, chains=4)
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Sampling and diagnostics

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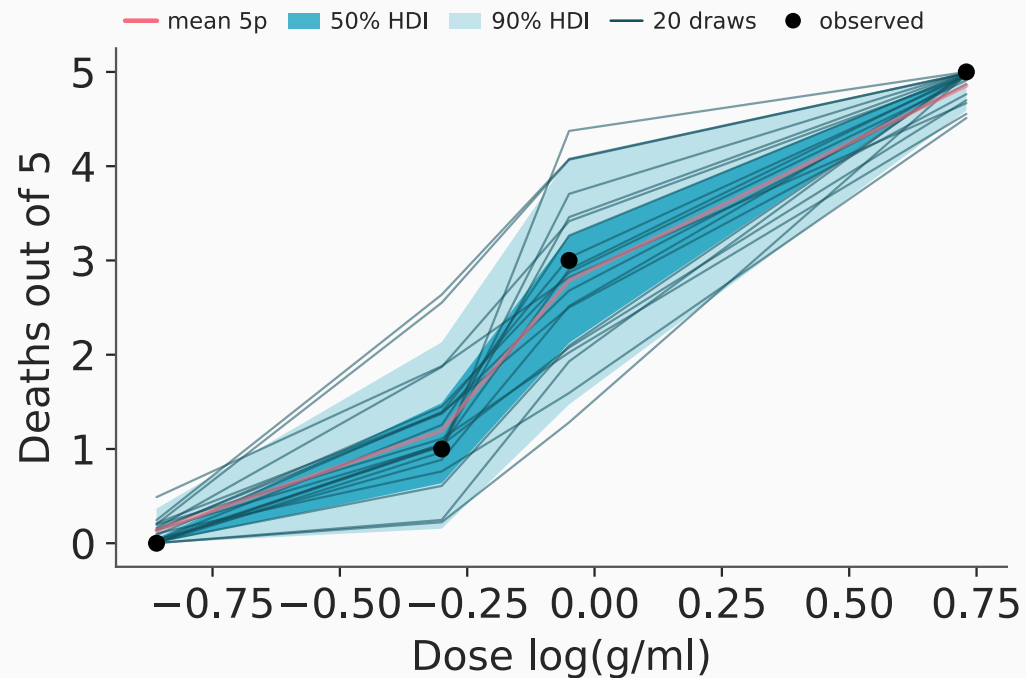


Check before interpreting:

- 0 divergences
- $\hat{R} = 1.00$
- ESS > 1,300 of 4,000 draws

Posterior: which dose-response curves remain plausible?

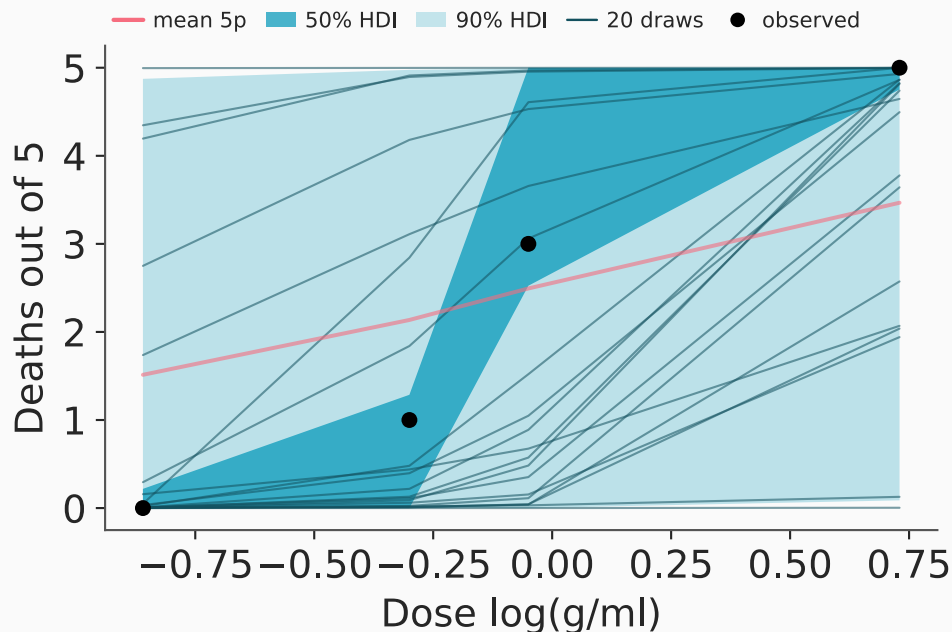
The bands show uncertainty about the mean number of deaths.



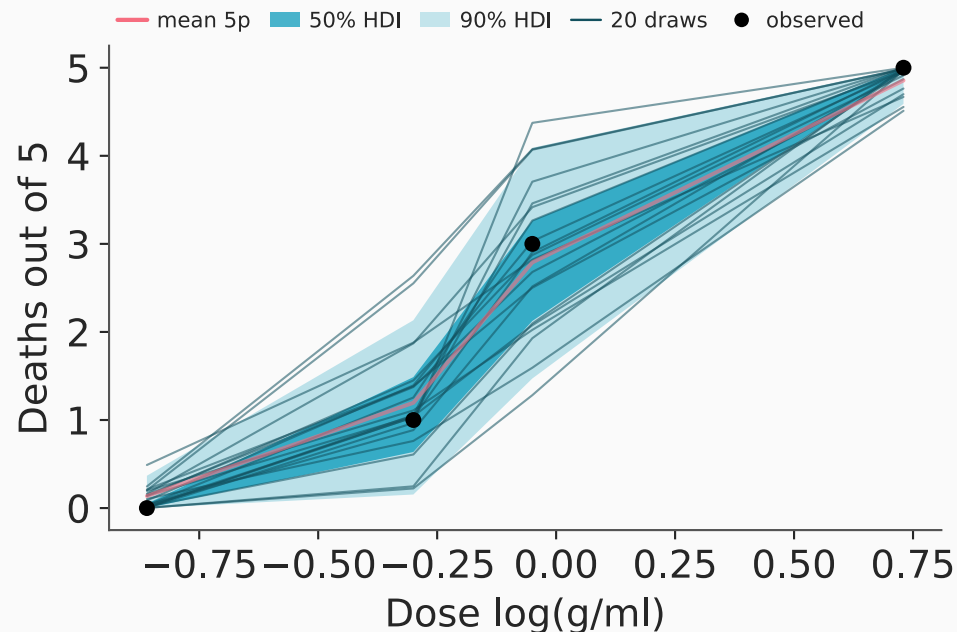
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Prior

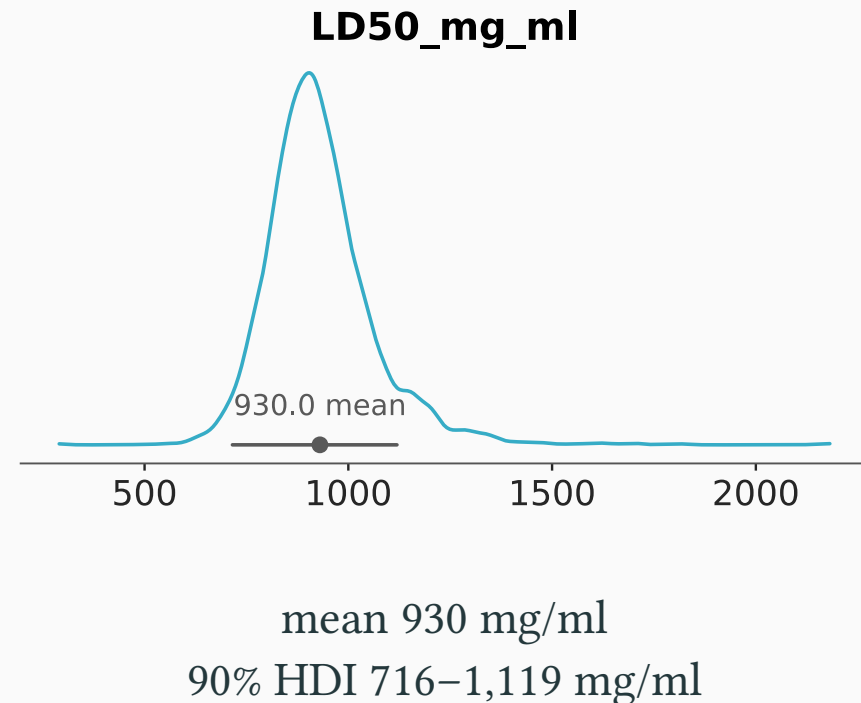


Posterior



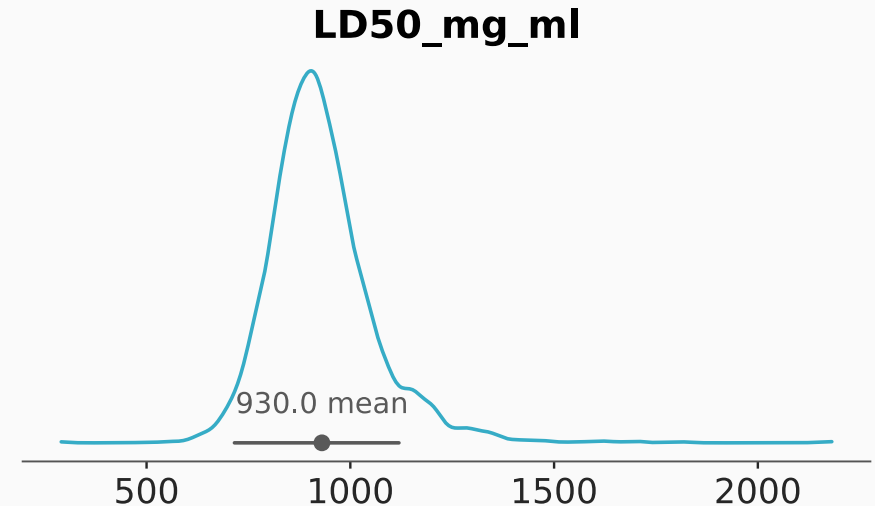
LD50: the quantity we actually care about

- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.



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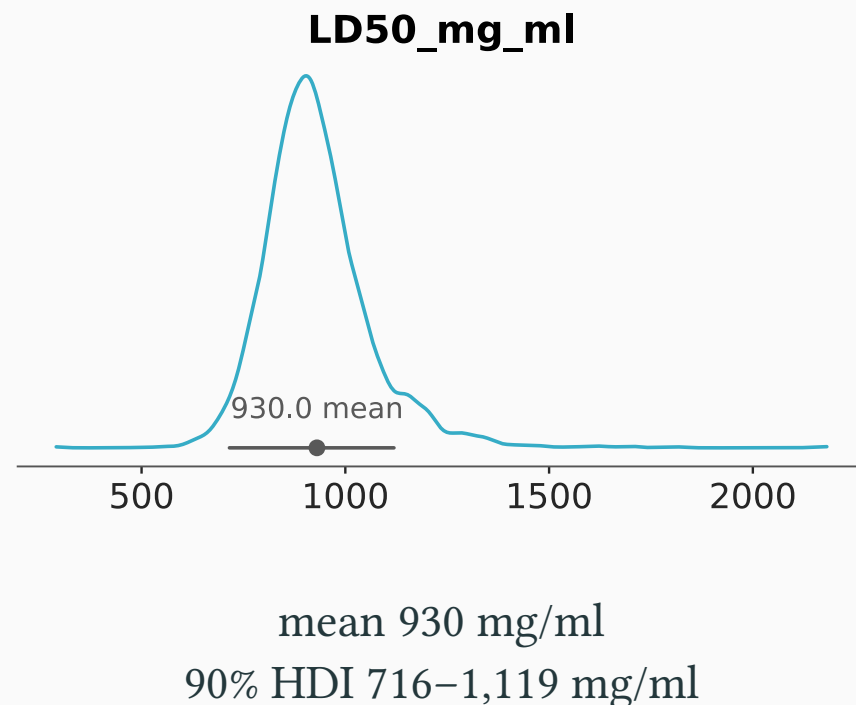
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 - $LD50 = -\alpha/\beta$
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- So every quantity comes with its own uncertainty.



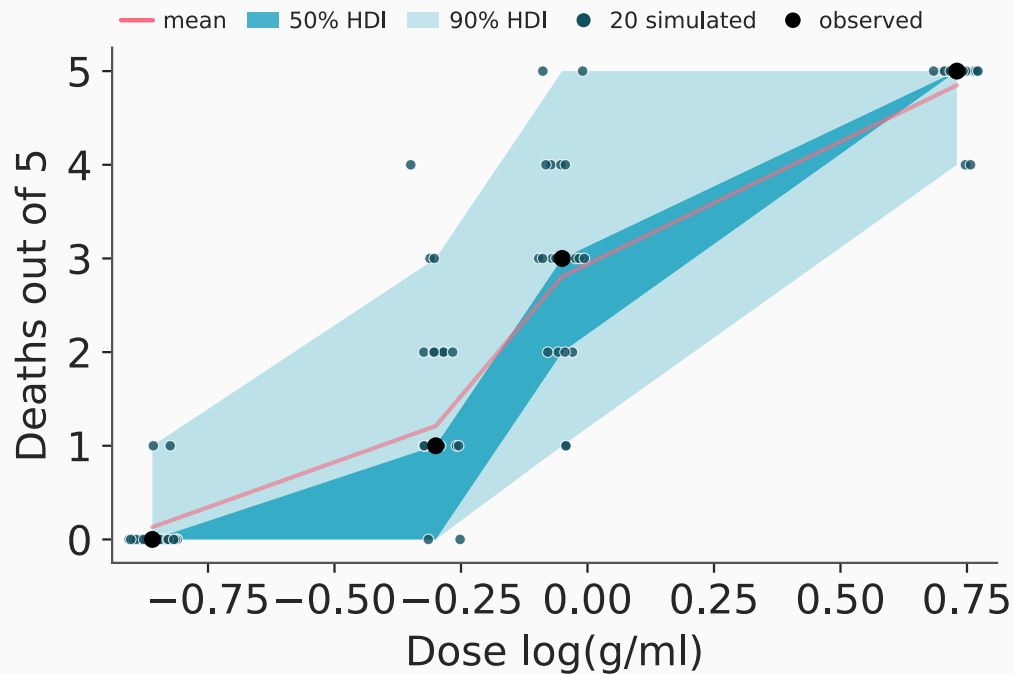
mean 930 mg/ml
90% HDI 716–1,119 mg/ml

LD50: the quantity we actually care about

- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.
- So every quantity comes with its own uncertainty.
- Here *how well* we know LD50 matters as much as its value.

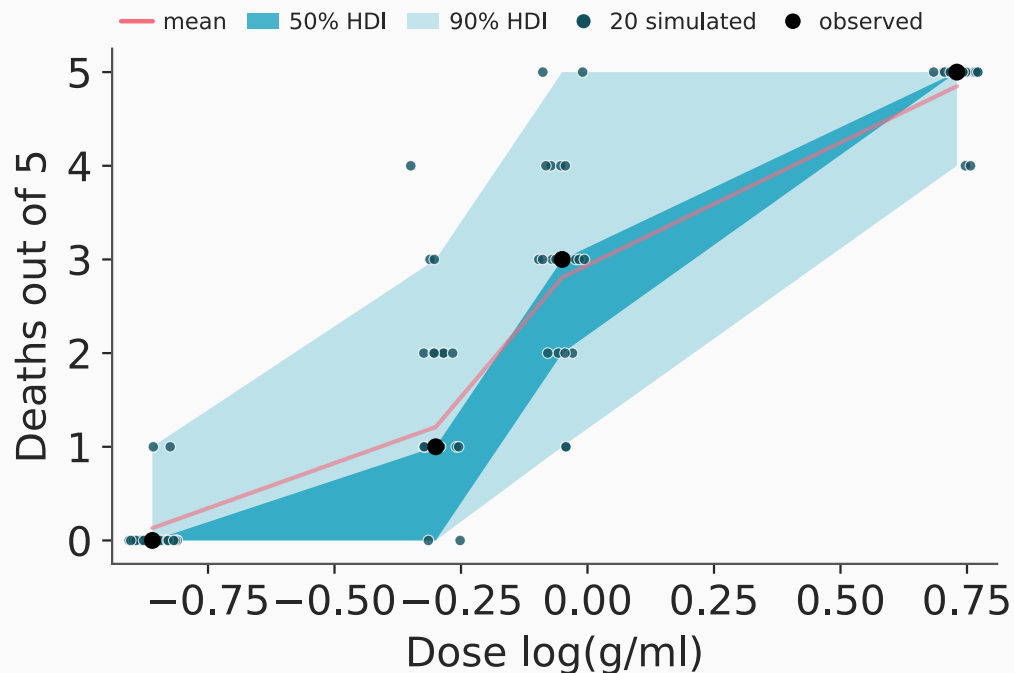


Posterior predictive: can the model reproduce the experiment?



This is what the model predicts we could have seen, *after* fitting.

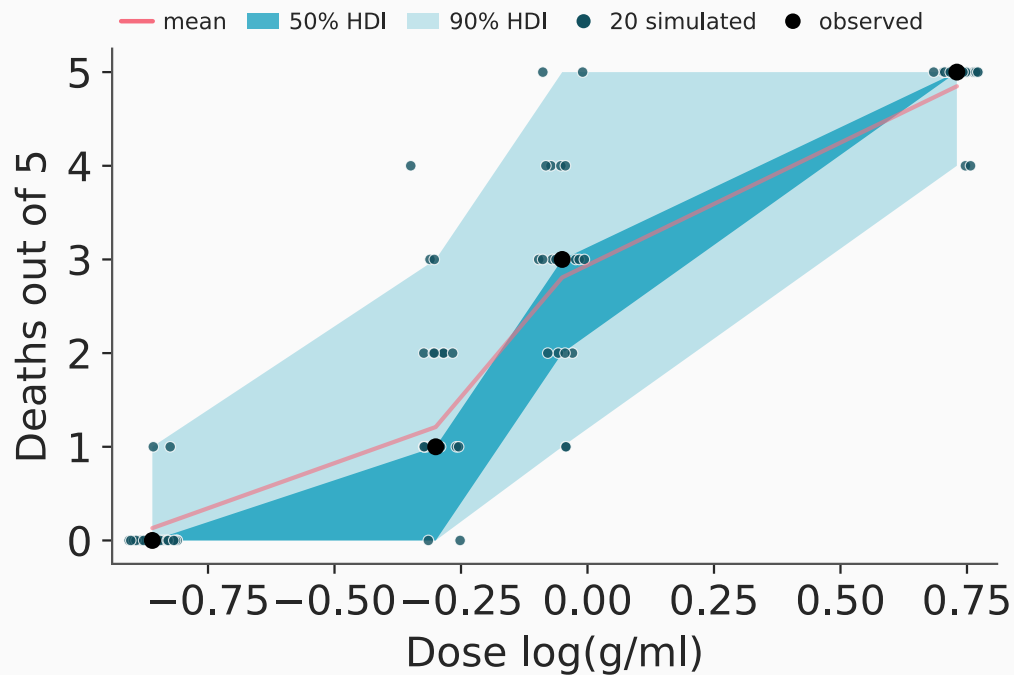
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- A workflow check meant to be done before looking at the posterior.

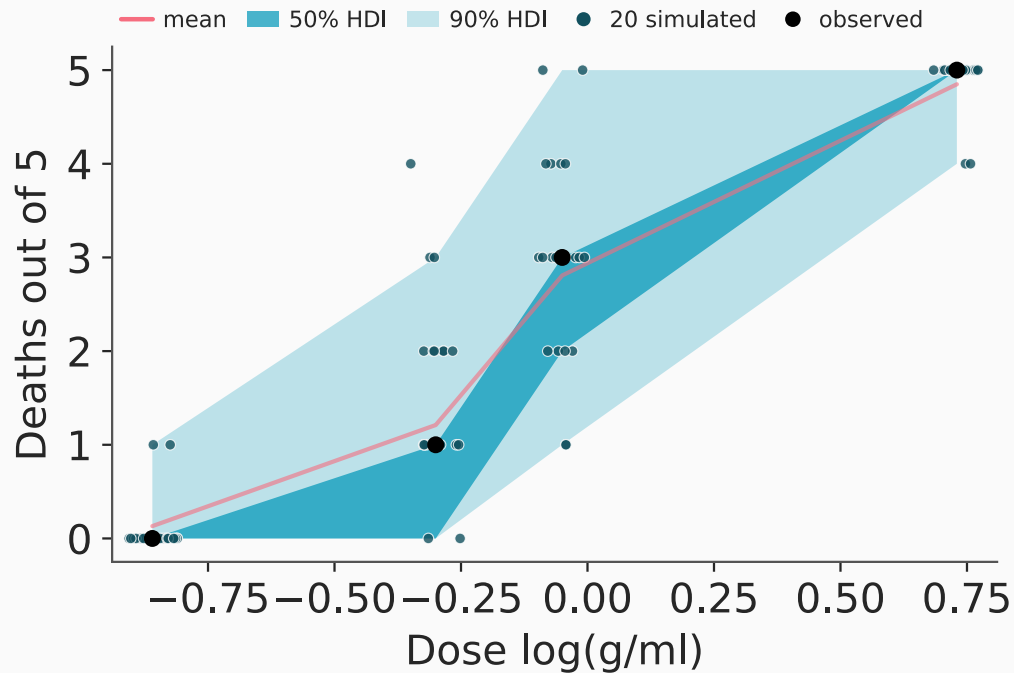
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- Workflow is not lockstep: going back to check is always fine.

Posterior predictive: can the model reproduce the experiment?



This is what the model predicts we could have seen, *after* fitting.

- A workflow check meant to be done before looking at the posterior.
- Workflow is not lockstep: going back to check is always fine.
- We understand data better than we understand parameters.

The Bayesian toolkit

Four libraries, one workflow



Write the model, sample the posterior

pymc.io



Choose and check priors

preliz.readthedocs.io



Diagnose, summarize, and plot

python.arviz.org



Regression models from a formula

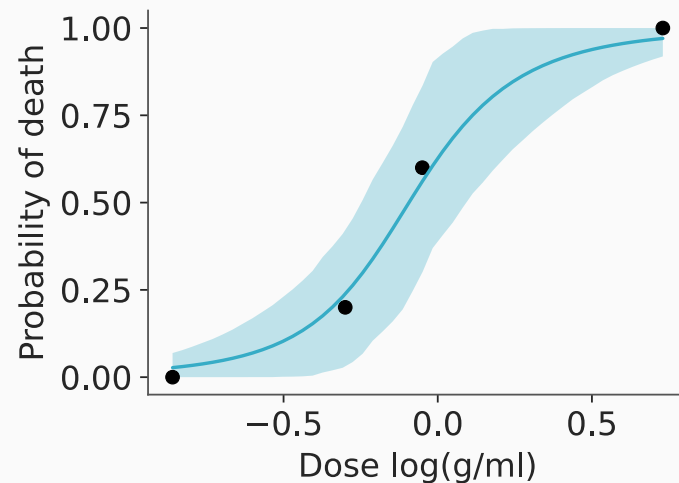
bambinos.github.io/bambi

Bambi: the same model as one formula

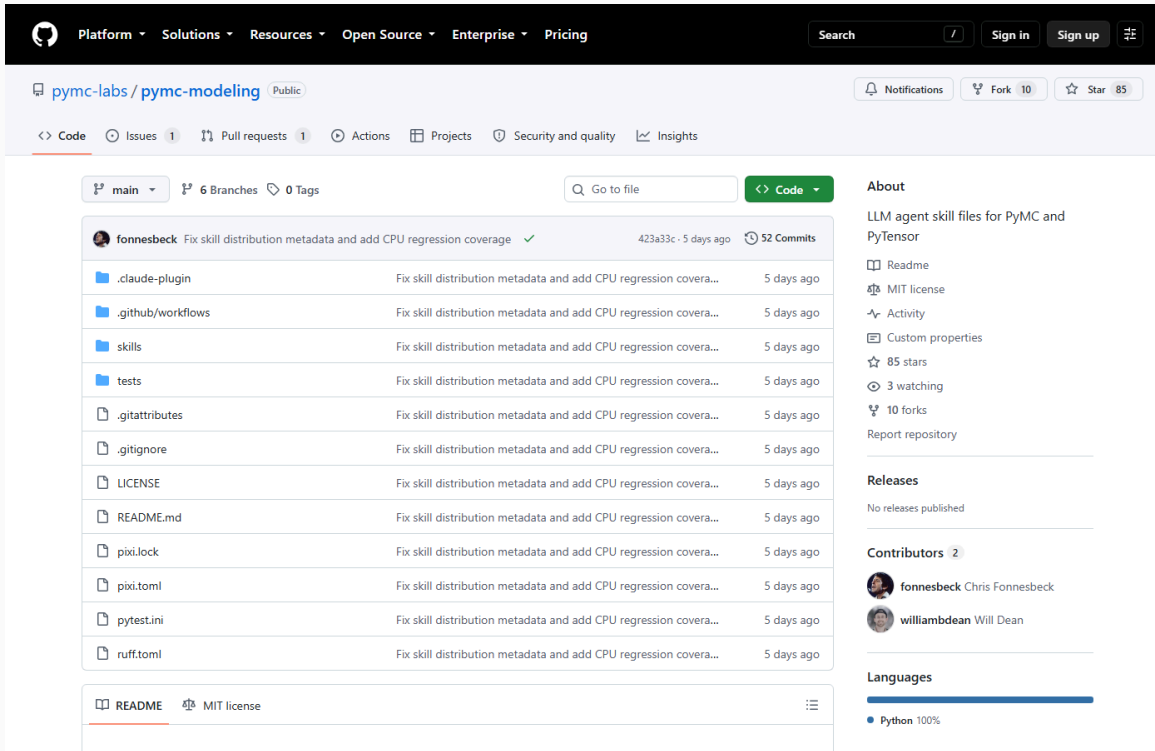
```
priors = {  
    "Intercept": bmb.Prior("Normal", mu=0, sigma=5),  
    "dose": bmb.Prior("HalfNormal", sigma=5),  
}  
model = bmb.Model("p(deaths, n) ~ dose", data,  
                  family="binomial", priors=priors)  
idata = model.fit()  
bmb.interpret.plot_predictions(model, idata, "dose")
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Coding assistants can learn current PyMC practice

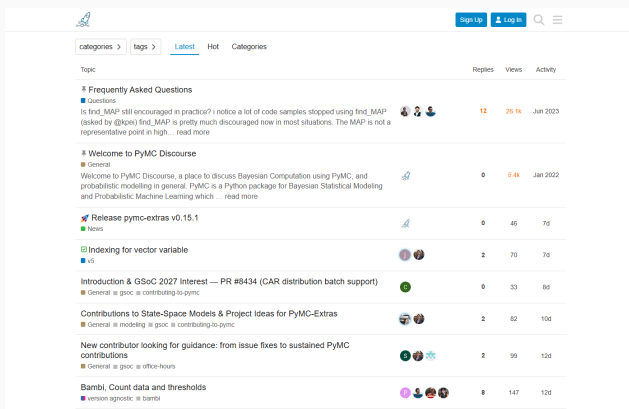


github.com/pymc-labs/pymc-modeling

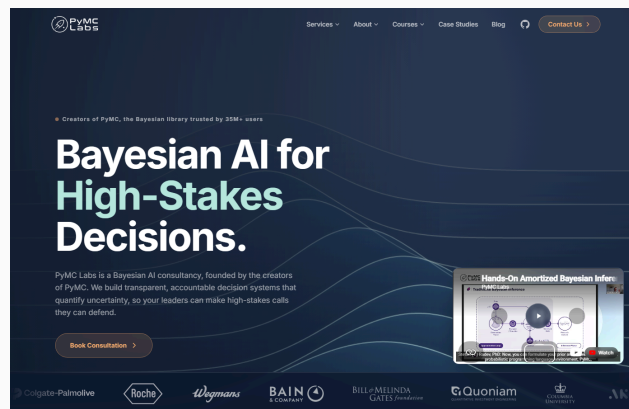
Skill files from PyMC Labs that teach AI coding assistants current PyMC and ArviZ practice.

We used them to build this course's notebooks.

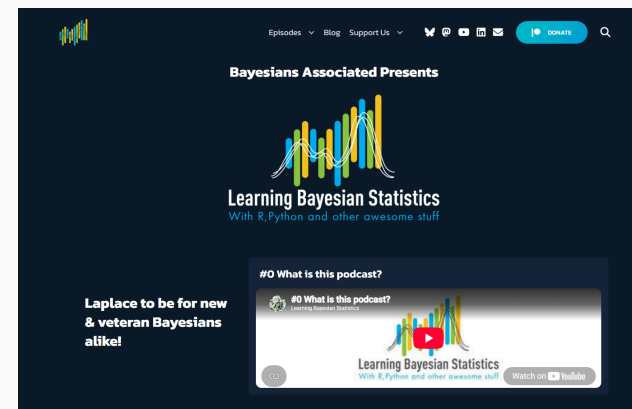
Where to ask, hire, and listen



Ask: PyMC Discourse
discourse.pymc.io



Hire: PyMC Labs
pymc-labs.com



Listen: Learning Bayesian Statistics
learnbayesstats.com

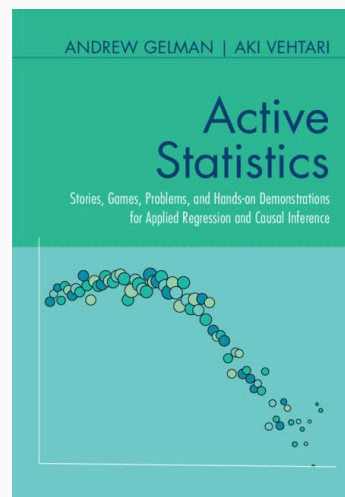
Books worth reading next



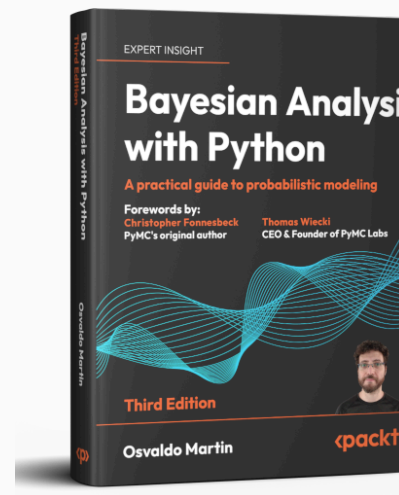
McElreath
Statistical Rethinking



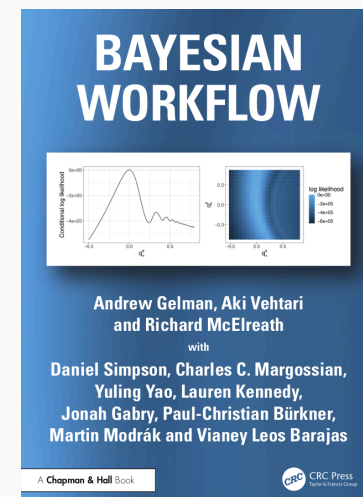
Gelman, Hill & Vehtari
Regression and Other Stories



Gelman & Vehtari
Active Statistics



Martin
Bayesian Analysis with Python



Gelman, Vehtari et al.
Bayesian Workflow

The Bayesian workflow

The workflow starts with a four-step loop



Data

The workflow starts with a four-step loop



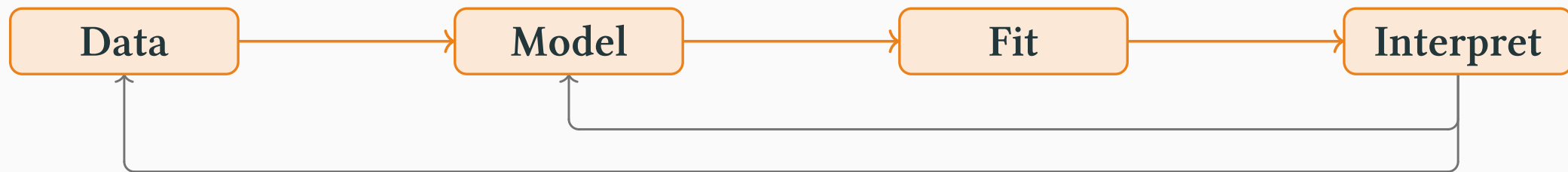
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Even at this level, we allow for

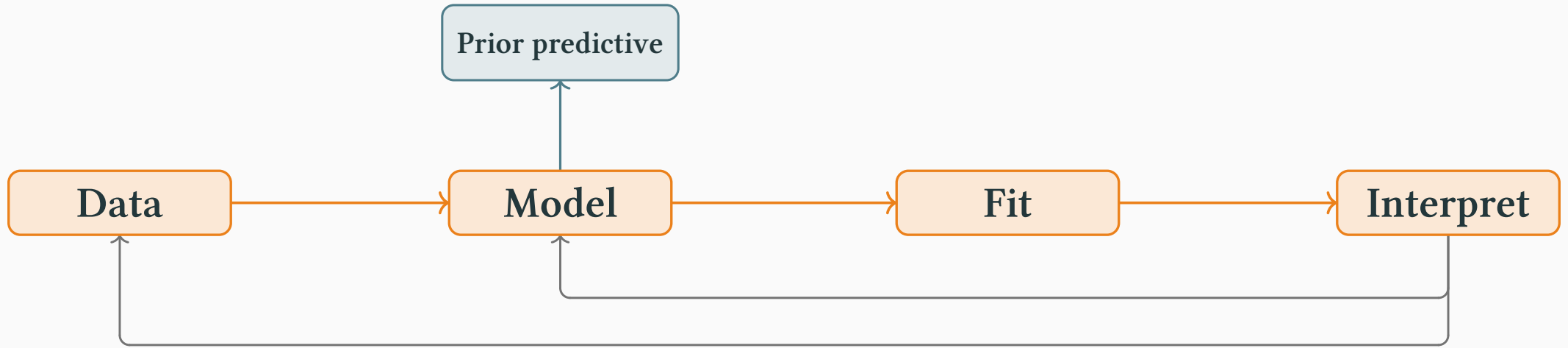
different interpretations · iterative experimental design

Checks make each transition accountable



Validation belongs at each step

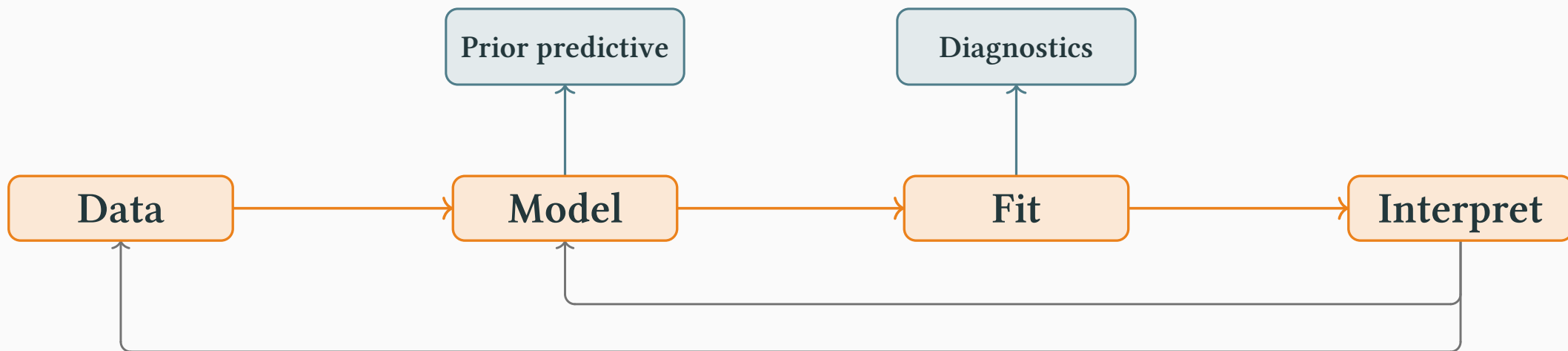
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Prior predictive

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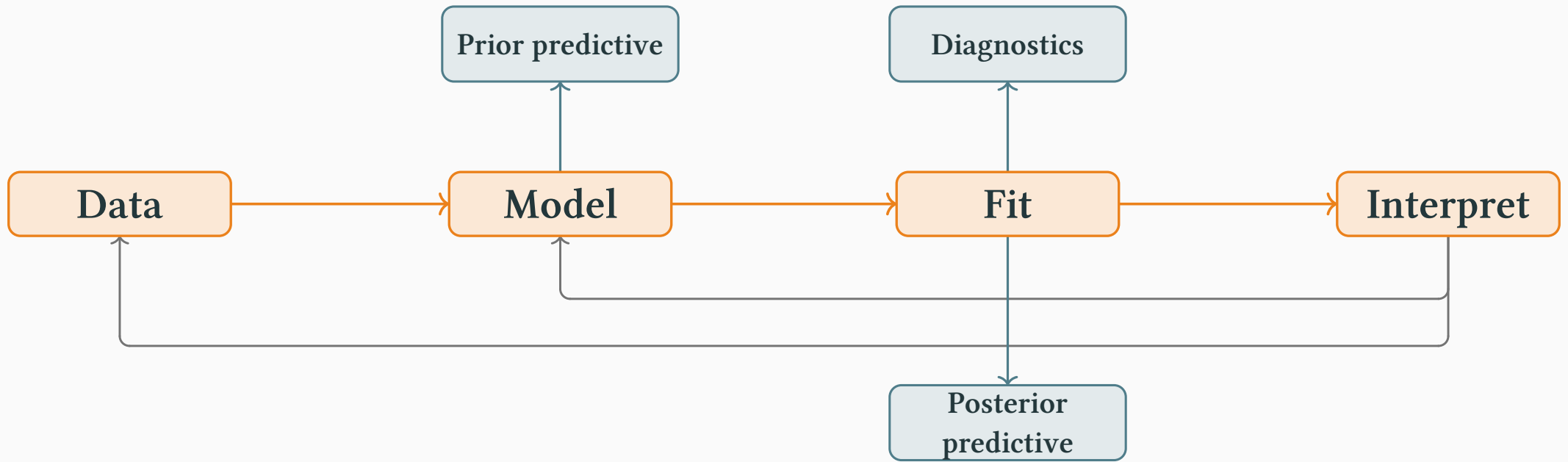
Prior predictive

Diagnostics

tests model coherence

test whether fitting converged

Checks make each transition accountable



Validation belongs at each step

Prior predictive

Diagnostics

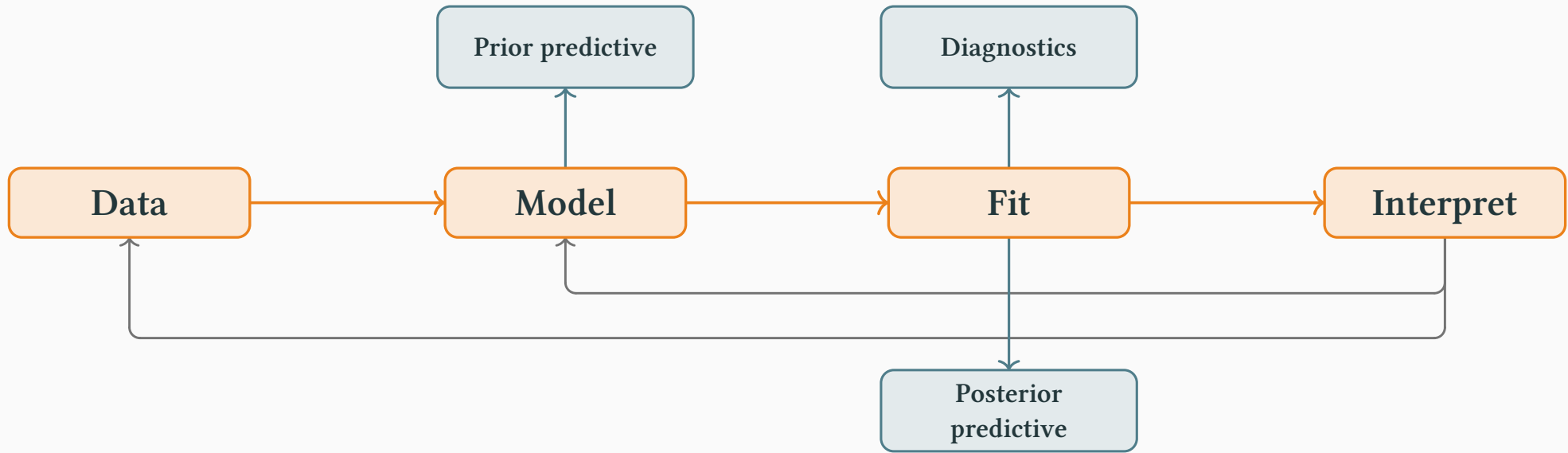
Posterior predictive

tests model coherence

test whether fitting converged

tests fit to the observed data

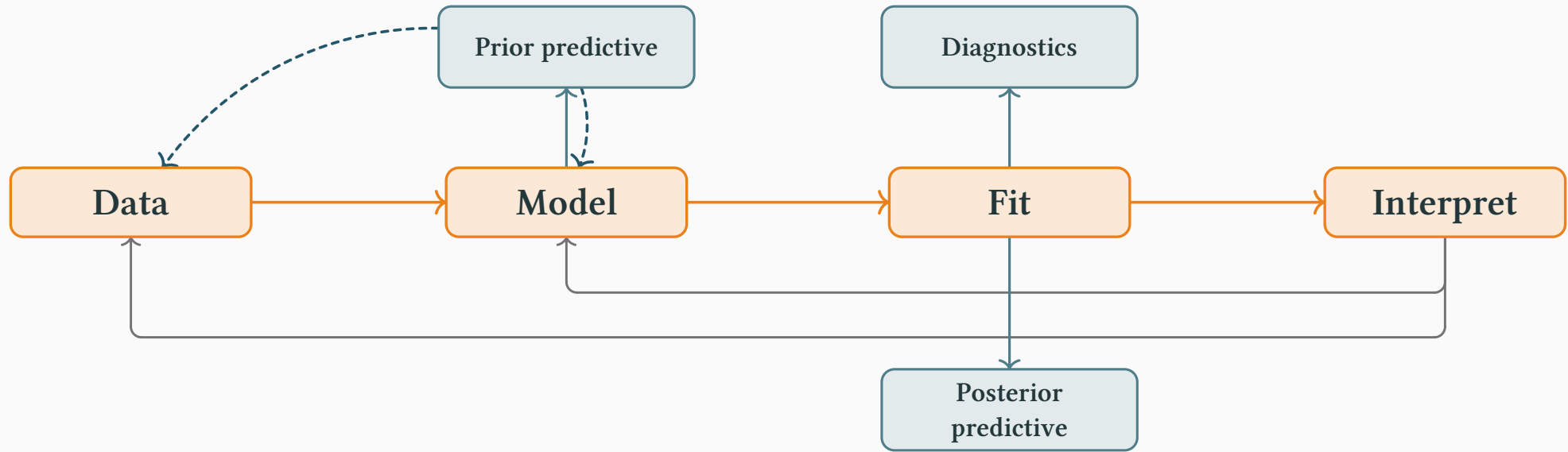
A failed check tells us where to return



A failed check

means something must change

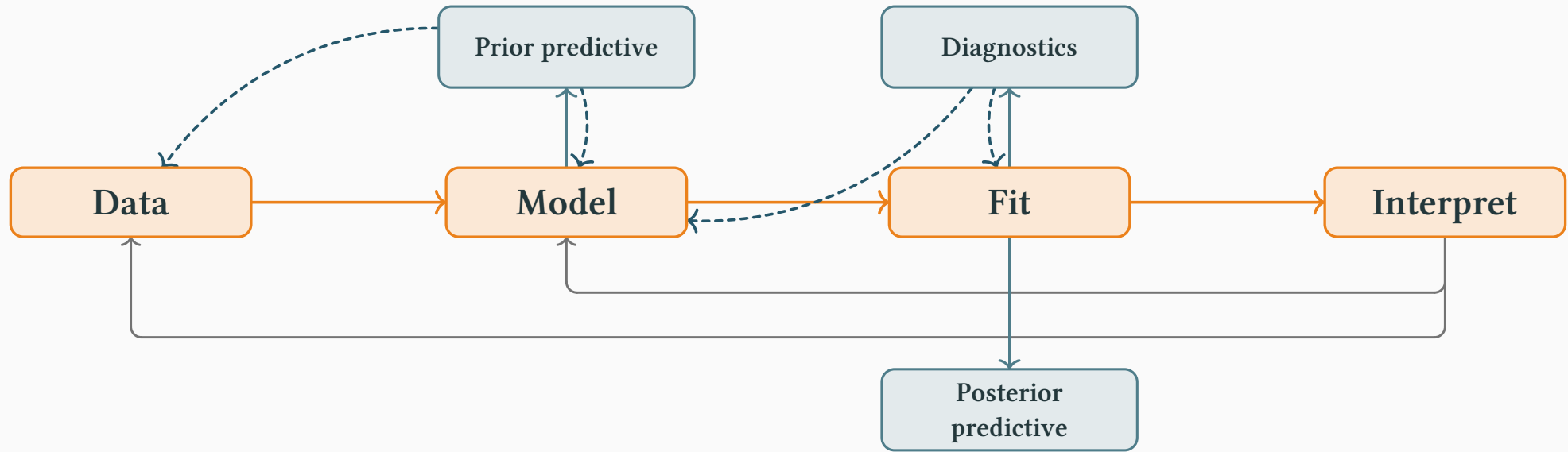
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Prior predictive

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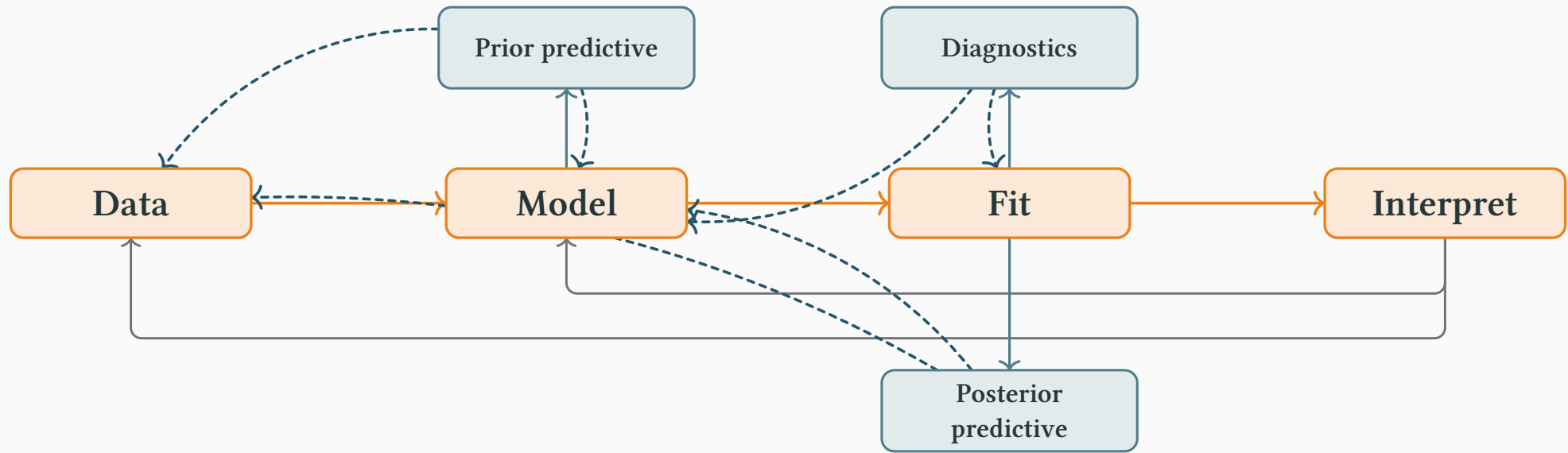
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A failed check
Prior predictive
Diagnostics

means something must change
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A failed check tells us where to return



A failed check
Prior predictive
Diagnostics
Posterior predictive

means something must change
revisit assumptions about the model or data
revisit the model or the computation
reassess how the model represents the data

Working through the workflow: golf putting

Preparing for active learning: sleep deprivation
